

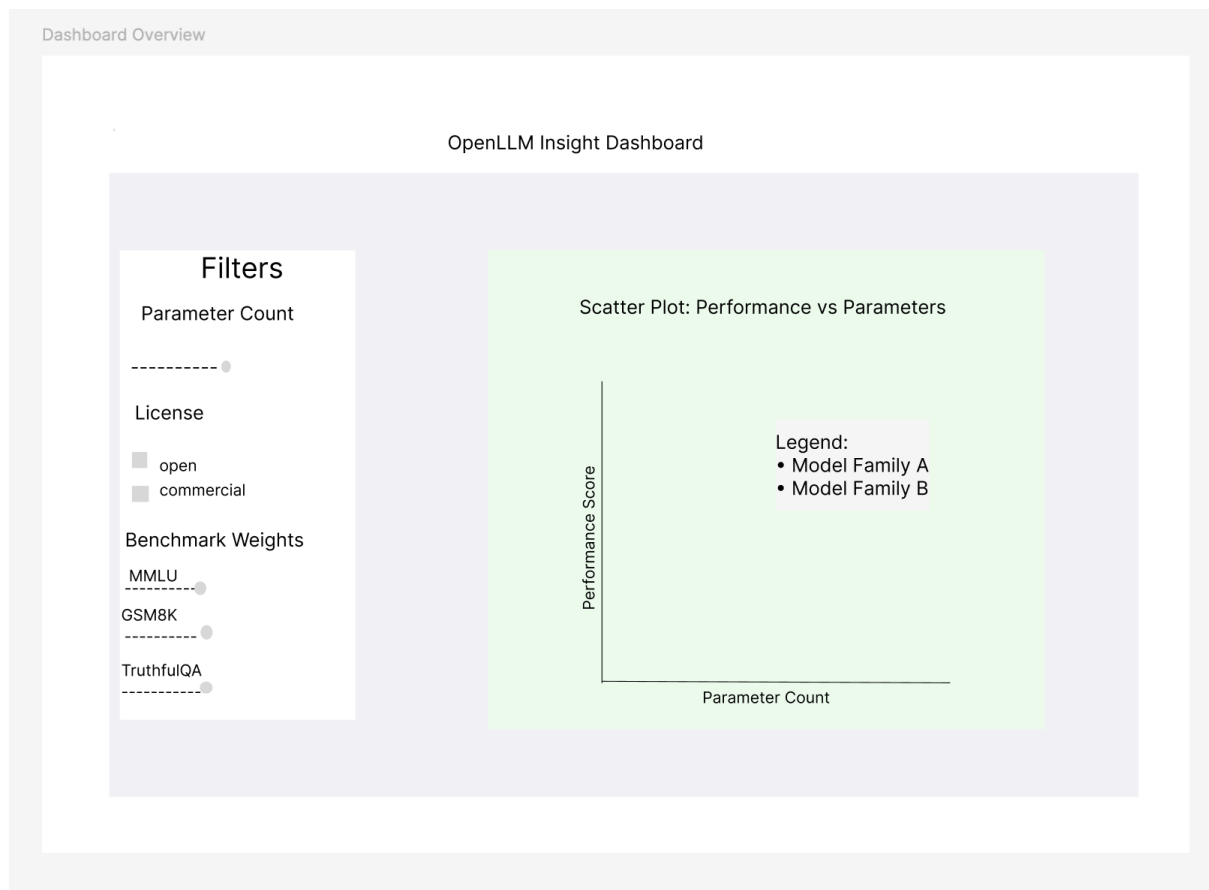
1. **Group ID:** Group 3

Group Members: Enrique Cabral-Mixco, Hemasri Bavisetty, Nipun Taneja

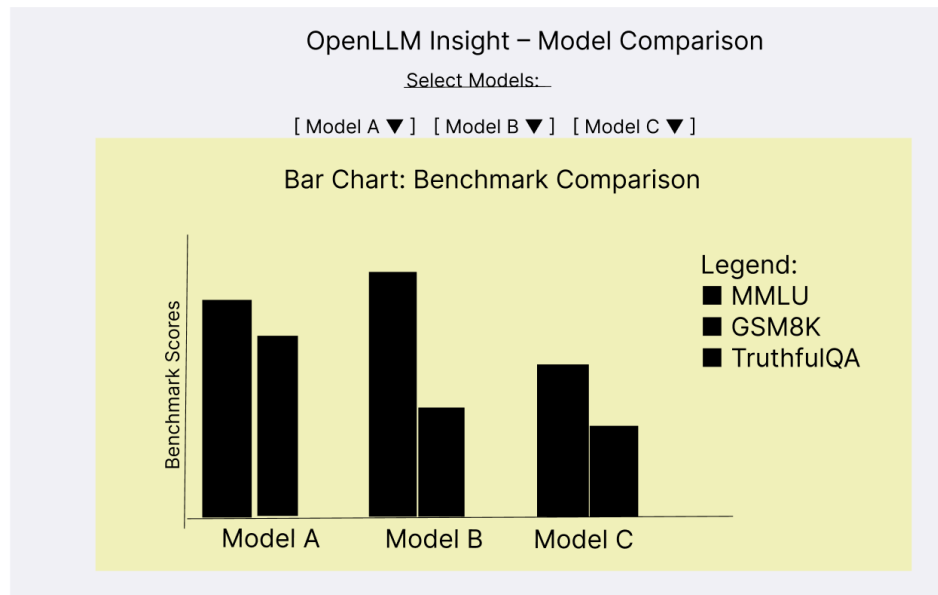
2. **Project Name:** OpenLLM Insight: A Visual Model Comparison System for LLM

Wireframe: Submit your wireframe or give online access to your wireframe.

2. Report: A report containing all the screen-shots of your wireframes with a short description (2-4 sentences) of each screen shot.

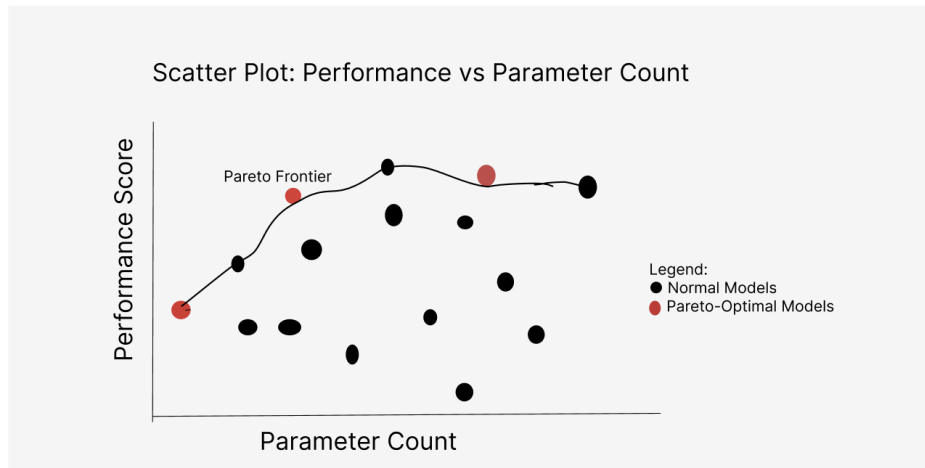


This screen presents the main dashboard of the OpenLLM Insight system. It visualizes models using a scatter plot where the x-axis represents parameter count and the y-axis represents performance score. Users can apply filters such as parameter count, license type, and benchmark weights to dynamically explore and analyze different models. The legend helps distinguish between different model families.

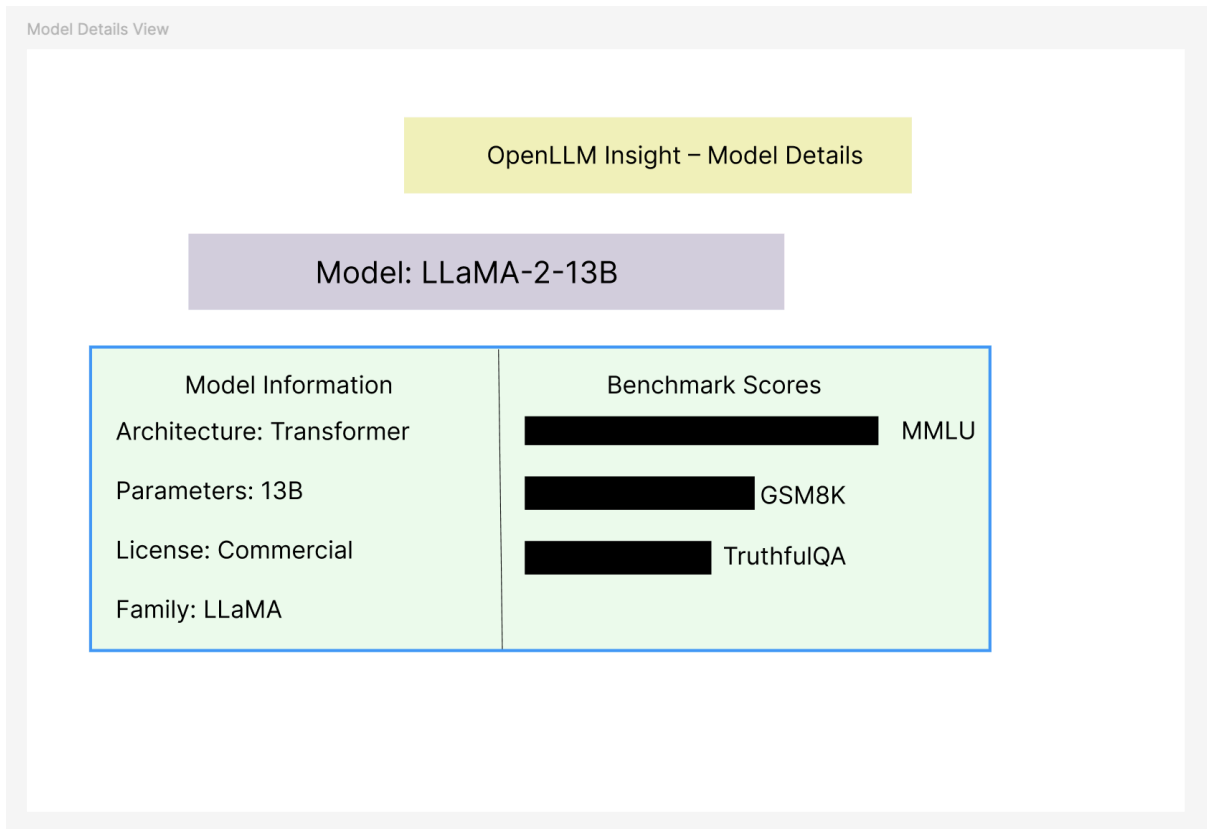


This screen allows users to compare multiple models side-by-side. Users can select models using dropdown menus and view their performance across benchmarks through a bar chart. The chart displays benchmark scores such as MMLU, GSM8K, and TruthfulQA, enabling users to analyze strengths and weaknesses of each model. A legend is included to identify different benchmarks.

OpenLLM Insight – Pareto Analysis



This screen visualizes the trade-off between model performance and parameter count using a scatter plot. Each point represents a model, and the Pareto frontier is highlighted to identify optimal models that provide the best performance for a given size. Pareto-optimal models are visually distinguished from other models, allowing users to quickly identify efficient choices for deployment.



This screen provides detailed information about a selected model. It includes metadata such as architecture, parameter count, license type, and model family. Additionally, a bar chart displays benchmark scores, offering a breakdown of the model's performance across different evaluation metrics. This view enables users to gain deeper insights into individual model characteristics.

Technologies

The selected platform for this project is a web-based visualization system. A web-based approach was chosen because it allows interactive exploration, easy access through a browser, and flexibility in designing multiple linked views such as dashboards, comparison charts, and detailed model pages. This approach is more suitable for the project than static tools because users need to filter models, compare benchmark scores, and inspect Pareto-optimal models dynamically.

For the frontend, React.js will be used to build the user interface. React is suitable because it supports reusable components and makes it easier to organize screens such as the dashboard, model comparison view, Pareto analysis view, and model details view. For visualization, D3.js will be used to create interactive charts such as scatter plots and bar charts. D3.js is a powerful visualization library that provides flexibility in customizing axes, legends, model points, and interactive behavior.

For data handling and preprocessing, Python with Pandas will be used. Pandas makes it easy to clean the dataset, remove unnecessary values, standardize parameter and score formats, and prepare the final CSV file for visualization. The dataset itself will be stored in

CSV format, since CSV is portable, easy to process, and integrates well with both Python and web-based applications.

The main technologies selected for this project are:

- **Platform:** Web-based system
- **Frontend:** React.js
- **Visualization Library:** D3.js
- **Data Processing:** Python (Pandas)
- **Data Format:** CSV

These technologies were selected because they are practical, widely used, and suitable for building an interactive visualization system for comparing large language models.